

Lesson Content for Ring & Run STEM Camp (2025)

Project Title: *Build a Smart AI-Powered Doorbell with Raspberry Pi*

Format: 3-Day, Hands-On Learning

Day 1: Raspberry Pi Setup & Hardware Prototyping

◆ Module 1: Configure the Raspberry Pi

Lesson Content:

In this module, students begin by preparing the Raspberry Pi 5 to act as the core of the smart doorbell system. The process starts by using the Raspberry Pi Imager tool to flash the latest Raspberry Pi OS (64-bit) onto a microSD card. Students configure essential settings such as hostname (based on team name), Wi-Fi credentials, username/password, and locale preferences.

After flashing the OS, students insert the SD card into the Raspberry Pi, connect a display and keyboard, and power it up. They learn how to use basic Linux commands to identify their device's IP address and verify network connectivity. By enabling SSH and VNC, students unlock the ability to remotely access and control their Pi—laying the foundation for hands-on development and debugging over the network.

Key learning points include:

- Understanding the purpose and capabilities of Raspberry Pi
 - Introduction to the Linux command line
 - Importance of IP addresses and remote access in IoT projects
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◆ Module 2: Assemble the Doorbell Hardware

Lesson Content:

In this hands-on module, students build a functioning doorbell prototype by assembling real electronic components. Using a breadboard and a T-Cobbler GPIO adapter, they connect a PIR (motion) sensor, push button, LED, resistor, and the Pi Camera Module to the Raspberry Pi.

Students gain a practical understanding of how electronic signals are transferred through GPIO pins. They follow a Fritzing diagram to carefully wire each component and troubleshoot

common hardware setup mistakes. Once powered, they verify the functionality by testing motion detection (PIR sensor) and LED response.

Key learning points include:

- Understanding breadboard layout and how it supports rapid prototyping
 - Basic principles of input/output pins, resistors, LEDs, and sensors
 - Interpreting pin diagrams and translating them into working circuits
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◆ **Module 3 (Part 1): Install and Configure the Software**

Lesson Content:

Students now shift focus from hardware to software. They clone the smart doorbell project from GitHub and run a setup script that installs required packages, sets the Bluetooth speaker, and enables necessary services.

Through this, students learn about Linux file permissions, package management, and what it means to “execute” a script. They also explore how Bluetooth devices are configured on Raspberry Pi and how volume/output preferences are set programmatically. This foundation prepares them to launch and control their device via code.

Key learning points include:

- Basics of Git/GitHub version control
 - File execution and Linux permissions using `chmod`
 - How setup scripts automate system preparation
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◆ **Module 3 (Part 2): Explore the Web App & MQTT Communication**

Lesson Content:

Here, students explore the structure of the software powering the doorbell. They distinguish between the front-end (user interface) and back-end (device control and communication). The front-end is written using HTML5, JavaScript, and CSS, while the back-end is a Python server that uses MQTT messaging to communicate with the browser.

Students learn to open and modify files in Visual Studio Code and test the client web app locally by entering the Raspberry Pi’s IP address in the browser. This module introduces real-time web communication and helps students understand how MQTT topics are used to pass messages between systems.

Key learning points include:

- The role of HTML, JavaScript, and CSS in building user interfaces
 - MQTT publish/subscribe model in IoT projects
 - How front-end and back-end software communicate over a network
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Day 2: Secure Communication & AI Integration

◆ Module 4: Add TLS/SSL Security

Lesson Content:

In this module, students dive into cybersecurity by securing the communication between the web app and the Raspberry Pi. They generate digital certificates using a shell script and configure the Mosquitto MQTT broker to use TLS encryption.

Students import certificates into their web browser to establish trust and ensure secure communication over HTTPS. This prevents eavesdropping and unauthorized control of the device. They learn the difference between public and private keys, and the critical importance of encryption in IoT systems.

Key learning points include:

- TLS/SSL encryption and the purpose of digital certificates
 - How to generate certificates using OpenSSL
 - Using PKI to establish trust between devices and browsers
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◆ Module 4 (Cont.): Secure MQTT Communication

Lesson Content:

Now that certificates are in place, students reconfigure the MQTT broker to support encrypted messaging. They apply secure listeners, update config files, and restart services. This ensures that all device-to-web communication is protected against network-based attacks.

Students also discuss how real-world systems implement authentication in addition to encryption, and what happens when security is misconfigured.

Key learning points include:

- MQTT over TLS (port 8883) vs unsecured (1883)
- Restarting Linux services and validating system behavior
- Real-world implications of weak security in IoT environments

◆ Module 5: Add GPT-4o AI Integration

Lesson Content:

The smart doorbell now becomes truly “smart.” In this module, students use the OpenAI API to add AI image recognition. The system captures an image from the camera when prompted and sends it to GPT-4o via an API request. GPT returns a human-readable description of what the doorbell sees.

Students retrieve a hidden API key using a steganography challenge, decode it, and add it to the project’s `.env` file. They then edit the `ring_server.py` backend file to enable and test the GPT integration.

Key learning points include:

- What is an API and how it enables applications to talk to AI
- How GPT-4o processes images and returns descriptive responses
- Importance of protecting API keys like passwords

Day 3: Customization & Project Showcase

◆ Module 6: Doorbell Customization Projects

Lesson Content:

With a fully functional secure smart doorbell in place, students now explore ways to make it their own. They can change the web interface appearance, add new doorbell ringtones, upload recorded footage to cloud storage, or experiment with different AI prompts.

This module encourages creativity, problem-solving, and teamwork. Students apply what they've learned and think critically about enhancements that improve user experience or security.

Key learning points include:

- How software customization creates user-focused experiences
- Exploring cloud services like AWS S3 or Azure Blob Storage
- Applying CSS, Python, or MQTT knowledge in new contexts

◆ Module 7: Team Presentations & Reflection

Lesson Content:

To close the camp, each team presents their final project. They describe what they built, how they customized it, what they learned, and any challenges they faced. This builds confidence, encourages peer learning, and reinforces key technical and communication skills.

Students also reflect on future learning paths in technology, cybersecurity, AI, and engineering. Resources are shared for continuing exploration, and one team member wins the Raspberry Pi Starter Kit prize!

Key learning points include:

- Presenting technical work clearly to a group
- Reflecting on technical and personal growth
- Identifying future opportunities in STEM